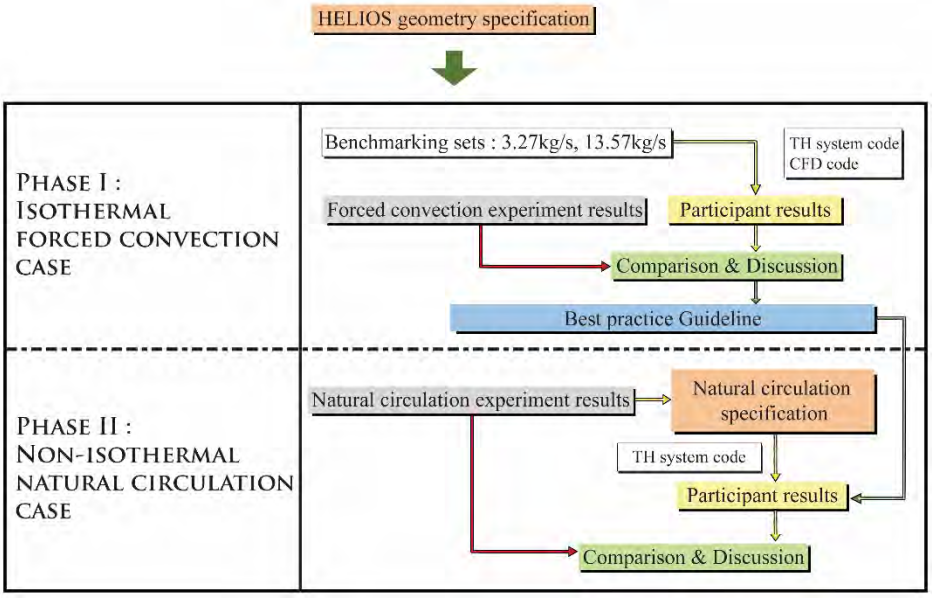
Figure 15.2.1: Overall procedure of NEA LACANES benchmarking



In Phase I, predictions of pressure losses calculated by programme participants were compared with measured data gathered from experimental results and supplementary information with other predictions. Participants used their own system codes to predict the pressure losses for all components of the HELIOS flow line. Furthermore, two CFD codes were simulated to obtain the pressure losses for specific components that have complicated geometries. The participants calculated pressure losses for two mass flow rate test conditions including low mass flow rate (3.27 kg/s) and high mass flow rate (13.57 kg/s) with an isothermal steady state of 250°C. Based on comparisons and discussions of benchmarking results, best practice guidelines to obtain the pressure loss coefficients on LACANES were constructed.

15.2.2 *Description of LBE thermo-hydraulic test loop HELIOS*

HELIOS was designed by scaling down with 5 000:1 ratio from its prototype PEACER-300 at Seoul National University. It has operated since May 2005 with the principle purposes of thermo-hydraulic experimentation and materials corrosion testing under LBE dynamic conditions. In the area of thermo-hydraulics, the verification of natural circulation capability that demonstrates the loss of flow accident (LOFA) condition of PEACER-300 is the main target scenario of HELIOS. Figure 15.2.2 shows an overall schematic design of PEACER-300 and the HELIOS demonstration LBE test loop.

Figure 15.2.3 shows the thermo-hydraulic loop configuration of the HELIOS system (except the support structure and material corrosion test section). It is comprised of the mock-up core, expansion tank, heat exchanger, mechanical pump, orifice, several gate valves and pipe system. The mock-up core generates heat by using four electrical heaters arranged in a square lattice as a principal heat source, having a maximum 60 kW thermal capacity. A heat exchanger removes the heat of LBE by flowing thermo-oil instead of water with counter-current flow between two fluids. In order to test the natural circulation capability of PEACER-300, the elevation difference between the heat source (a mock-up core) and the heat sink (a heat exchanger) is kept the same as that of PEACER-300, at about 7.4 m with a total height of 12 m.

Pressure losses and LBE temperatures are the main variables in the LACANES benchmarking study. Figure 15.2.3 also indicates four differential pressure metres[2] and ten thermocouples[3] installed at several regions. Four differential pressure metres have been installed in mock-up core, orifice, gate valve and a combined region that covers the area from core outlet to orifice outlet.

Figure 15.2.2: Design schematic of PEACER-300 and picture of demonstration LBE test loop, HELIOS

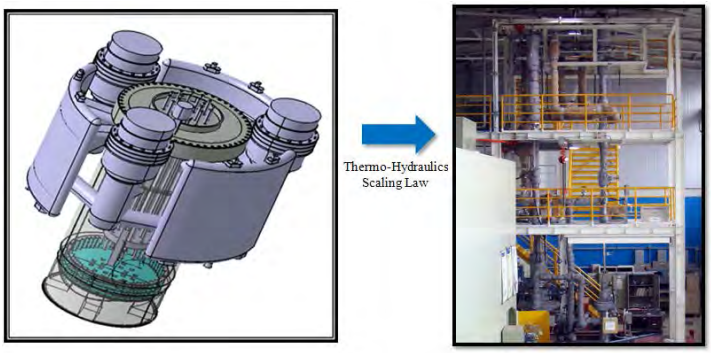
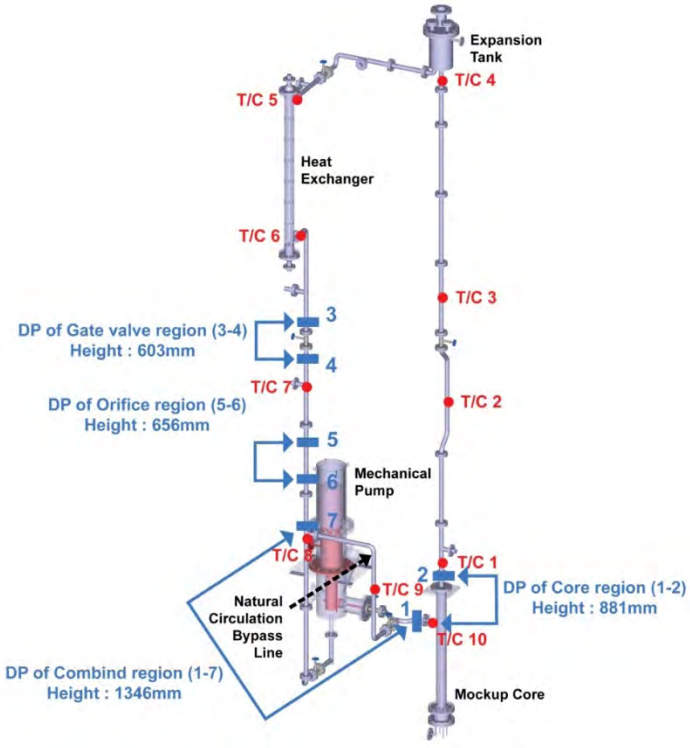


Figure 15.2.3: Entire loop configuration including several locations of differential pressure metres and thermocouples in HELIOS



2. Rosemount-3051 CD3A, Rosemount Company, 2.5 bar measurement span with error of ±0.065% full scale.
3. Type K with stainless steel 304, calibrated with ±0.5°C error.

In contrast the detailed modelling of the complex geometry using CFD tools produced results that agree well with HELIOS experimental data, as shown in the figure above. It has been found that CFD modelling always provided more accurate results than experimental correlations (Cho, 2011). Further examples are given in Section 15.3.6.

15.3.5 *Sudden area change*

In regions like the gate valve and connecting piece of flange section, flow areas suddenly change, which disturbs the flow. The sudden area changes are illustrated in Figure 15.3.6 and correlations are listed in Table 15.3.5.

Figure 15.3.6: Geometries of sudden area changes

Sudden expansion (left), sudden contraction (right)

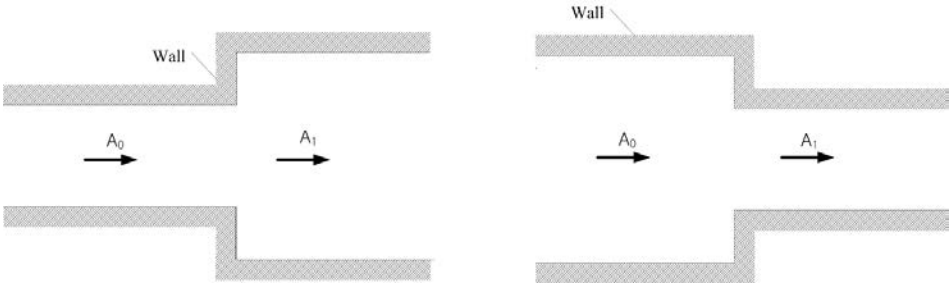


Table 15.3.5: Formulas for calculation of form loss coefficients in sudden expansion of flow area

Sudden expansion	Range	Reference
$K = \left(1 - \frac{A_0}{A_1}\right)^2$ Borda-Carnot's formula	$Re > 3.3 \cdot 10^3$	Idelchik (2000)

Table 15.3.6: Formulas for calculation of form loss coefficient in sudden contraction of flow area

	Sudden contraction	Range	Reference
1	$K = 0.5\left(1 - \frac{A_1}{A_0}\right)$	$Re > 10^4$	Idelchik (2000)
2	$K = 0.5\left(1 - \frac{A_1}{A_0}\right)^{3/4}$	$Re > 10^4$	Idelchik (1992)
3	$K = 0.5 - 0.7 \cdot \left(\frac{A_1}{A_0}\right) + 0.2 \cdot \left(\frac{A_1}{A_0}\right)^2$	$Re > 10^4$	Massey (1968)
4	$K = \left(1 - \frac{A_1}{A_c}\right)^2$ $\frac{A_c}{A_1} = 0.62 + 0.38\left(\frac{A_1}{A_0}\right)^3$	All range	Vennard (1961)